Să se determine următoarele formule de cuadratură cu grad maxim de exactitate

a) $\int_{-\infty}^{+\infty} e^{-x^2} f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + R(f)$

x_0, x_1 - noduri; A_0, A_1 - coeficienți; $n=1$

Gradul maxim care poate fi atins $= 2\cdot n+1 = 3$

$\Rightarrow R(P)=0 \quad \forall P \in \Pi_3 \Leftrightarrow R(1)=R(x)=R(x^2)=R(x^3)=0$

$w(x) = e^{-x^2}$ Este o formulă de cuadratură de tip Gauss-Hermite. Nodurile sunt rădăcinile polinomului Hermite de gradul 2, $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2})$

$H_2(x) = (-1)^2 e^{x^2} \frac{d^2}{dx^2}(e^{-x^2}) = 4x^2-2$

$4x^2-2=0 \Rightarrow x_0 = -\frac{\sqrt{2}}{2};\ x_1 = \frac{\sqrt{2}}{2}$

Pt a afla coeficienții A_0 și A_1 considerăm

$f=1 \Rightarrow A_0\cdot 1 + A_1\cdot 1 = \int_{-\infty}^{+\infty} e^{-x^2}dx = \sqrt{\pi}$

$f=x \Rightarrow A_0\cdot x_0 + A_1\cdot x_1 = \int_{-\infty}^{+\infty} x e^{-x^2}dx = 0$

$\Rightarrow \begin{cases} A_0+A_1=\sqrt{\pi} \\ \frac{\sqrt{2}}{2}(A_1-A_0)=0 \end{cases} \Rightarrow A_0=A_1=\sqrt{\pi}/2$

$\Rightarrow \int_{-\infty}^{+\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{2}\left(f\left(\frac{-\sqrt{2}}{2}\right)+f\left(\frac{\sqrt{2}}{2}\right)\right)+R(f)$

are gradul de exactitate 3

b) $\int_{-\infty}^{+\infty} e^{-4x^2} f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + R(f)$

$\int_{-\infty}^{+\infty} e^{-4x^2} f(x)\,dx = \int_{-\infty}^{+\infty} e^{-(2x)^2} f(x)\,dx = \frac{1}{2}\int_{-\infty}^{+\infty} e^{-t^2} f\left(\frac{t}{2}\right)dt$

$2x=t \Rightarrow x=t/2$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-4x^2} f(x)\,dx = \frac{1}{2}\int_{-\infty}^{+\infty} e^{-t^2} \underbrace{f(t/2)}_{=g(t)}\,dt = \frac{1}{2}\int_{-\infty}^{+\infty} e^{-t^2} g(t)\,dt$$

Problema se reduce la

$$\int_{-\infty}^{+\infty} e^{-t^2} g(t)\,dt = A_0' g(t_0) + A_1' g(t_1) + R(g)$$

$$\Rightarrow A_0' = A_1' = \frac{\sqrt{\pi}}{2}; \quad t_0 = -\frac{\sqrt{2}}{2}, \; t_1 = \frac{\sqrt{2}}{2}$$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-4x^2} f(x)\,dx = \frac{1}{2}\int_{-\infty}^{+\infty} e^{-t^2} g(t)\,dt =$$

$$= \frac{1}{2}\left(\frac{\sqrt{\pi}}{2} g\left(-\frac{\sqrt{2}}{2}\right) + \frac{\sqrt{\pi}}{2} g\left(\frac{\sqrt{2}}{2}\right) + R(g)\right) =$$

$$= \frac{\sqrt{\pi}}{4} f\left(-\frac{\sqrt{2}}{4}\right) + \frac{\sqrt{\pi}}{4} f\left(\frac{\sqrt{2}}{4}\right) + R(f)$$

$f(x) = f(t/2) = g(t)$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-4x^2} f(x)\,dx = \frac{\sqrt{\pi}}{4}\left(f\left(-\frac{\sqrt{2}}{4}\right) + f\left(\frac{\sqrt{2}}{4}\right)\right) + R(f)$$

are gradul de exactitate 3

c) $$\int_{-\infty}^{+\infty} e^{-x^2} f(x)\,dx = A_0 f(1) + A_1 f(x_1) + R(f)$$

A_0, A_1, x_1 se determină din $R(P) = 0 \; \forall P \in \Pi_2$

$(R(1) = R(x) = R(x^2) = 0)$

Fie $f = (x-1)(x-x_1)$ $\Rightarrow$ $f(1) = 0$, $f(x_1) = 0$

$$\Rightarrow \int_{-\infty}^{+\infty} (x-1)(x-x_1) e^{-x^2}\,dx = A_0 \cdot 0 + A_1 \cdot 0$$

$$\Rightarrow \int_{-\infty}^{+\infty} x(x-1) e^{-x^2}\,dx - x_1 \int_{-\infty}^{+\infty} (x-1) e^{-x^2}\,dx = 0$$

$$\Rightarrow x_1 = \frac{\int_{-\infty}^{+\infty} x(x-1) e^{-x^2}\,dx}{\int_{-\infty}^{+\infty} (x-1) e^{-x^2}\,dx} = \frac{\frac{\sqrt{\pi}}{2}}{-\sqrt{\pi}} = -\frac{1}{2}$$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-x^2} f(x)\,dx = A_0 f(1) + A_1 f\left(-\frac{1}{2}\right) + R(f)$$

Pt a determina coeficienții A_0 și A_1 considerăm

$$f=1 \Rightarrow A_0 + A_1 = \int_{-\infty}^{+\infty} e^{-x^2}dx = \sqrt{\pi}$$

$$f=x \Rightarrow A_0 \cdot 1 + A_1\left(-\frac{1}{2}\right) = \int_{-\infty}^{+\infty} x e^{-x^2} dx = 0$$

$$\Rightarrow \begin{cases} A_0 + A_1 = \sqrt{\pi} \\ 2A_0 - A_1 = 0 \end{cases} \Rightarrow \begin{cases} A_1 = 2A_0 \\ 3A_0 = \sqrt{\pi} \end{cases} \Rightarrow \begin{cases} A_0 = \frac{\sqrt{\pi}}{3} \\ A_1 = \frac{2}{3}\sqrt{\pi} \end{cases}$$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{3} f(1) + \frac{2\sqrt{\pi}}{3} f\left(-\frac{1}{2}\right) + R(f)$$

are gradul de exactitate 2